



CLS Linac Acceleration Section Upgrade Specifications

1.8.32.1- Rev. 0

Date: 2008-03-05

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Revision History

<i>Revision</i>	<i>Date</i>	<i>Description</i>	<i>Author</i>
A	2006-11-15	Original Draft	Xiaofeng Shen
B	2007-03-02	Draft-1	Xiaofeng Shen
C	2008-01-29	Version-1	Xiaofeng Shen
0	2008-03-05	Issued for Use	Xiaofeng Shen

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1.0 INTRODUCTION

1.1 PURPOSE

This document specifies the requirements for the design, fabrication, testing and delivery of four constant-gradient S-band linear acceleration structures for the CLS linac upgrade.

1.2 BACKGROUND

The operating linac in the CLS was established about forty years ago. Accelerating section #6 has developed some vacuum problems. And recently, some water leakage has occurred on the exterior water circuit. Sections #2, #3 and #4 are Varian type accelerating structures designed to have an operational lifetime of 20 years. However, these sections have been in operation for 45 years. And all are in a state of deterioration. It is therefore planned to replace all these four sections with SLAC type accelerating structures.

The S-band accelerating structures shall be the standard SLAC 3.05 meter, $2\pi/3$ mode and constant gradient disk-loaded cylindrical traveling waveguide structure. RF power unabsorbed in the section will be absorbed in an external water load mounted to the output coupler.

2.0 TECHNICAL SPECIFICATIONS

2.1 ELECTRICAL SPECIFICATIONS

- Type Constant gradient, traveling wave
- Structure Disk-loaded
- Operating mode $2\pi/3$ mode
- Operating frequency 2856.00 ± 0.1 MHz (in vacuum, 45°C)
- Operating temperature 45 ± 0.1 °C
- Peak RF input power ≥ 60 MW
- Pulse repetition rate 50 Hz
- RF pulse length (flattop) ≥ 3.0 μ s
- Input and output VSWR ≤ 1.08
- Bandwidth (VSWR ≤ 1.2) $\geq \pm 2$ MHz
- Phase velocity light velocity in vacuum
- RF period 3.499 cm
- Number of cavities 86 (including input and output couplers)
- Phase shift per cavity cell $120 \pm 2^\circ$
- Cumulative RF phase shift $(85 \times 120) \pm 2^\circ$
- Filling time 0.83 μ s
- Effective shunt impedance 53 to 60 M Ω /m
- RF attenuation parameter 0.57 neper
- Normalized group velocity 0.0204 to 0.0065
- Energy gain ≥ 25 MeV/m
- Structure quality factor $\geq 13,000$

The planned RF tuning procedure including temperature control shall be detailed by the seller and agreed upon by the buyer before fabrication begins.

2.2 MECHANICAL

2.2.1 Vacuum Leak Rate and Vacuum Pressure

- Vacuum tight leak rate $\leq 2 \times 10^{-10}$ Torr·liter/s of helium
- Vacuum pressure $\leq 5 \times 10^{-7}$ Torr @45°C with a 60l/s SIP pumping at the opposite end

2.2.2 Material

- The RF cavities shall be machined from OFE certified copper material (certified copies shall be transmitted to buyer), ASTM C10100 ^[1] or better [Cu + Ag > 99.994%, O < 0.0005%, Conductivity $\geq 58.6 \text{ m}/\Omega\cdot\text{mm}^2$].
- Water-cooling tubes shall be made of OF copper standard, ASTM C10200 ^[1] (Cu + Ag > 99.95%, O < 0.001%) or better.
- No magnetic materials are to be used for any part.

2.2.3 Waveguide

- The rectangular waveguide, WR-284 or equivalent, input and output connections shall be on the opposite sides of the section. For three of the structures, the input connection shall be on the right side as one looks from beam input side to output side. The fourth section input connection shall be on the left side as one looks from input to output.

2.2.4 Vacuum Flanges

- Stainless steel beam line vacuum flanges at each end of the structure shall be 6 inch standard “Conflat” flanges or equivalent unless otherwise specified by the buyer. Flanges shall be rotatable and shall be welded to stainless steel tubes, which are brazed to the outer walls of the coupler cavity. Flanges shall be mounted so that their faces are perpendicular to the structure axis within ± 0.1 miliradian.

2.2.5 RF Input/Output Couplers and Flanges

- Couplers will be built with “offset” (SLAC type). The vendor shall provide the necessary data to demonstrate field asymmetry compensation at the input and output coupling cells.
- The input side shall use SLAC type (Skarpass) female waveguide flange.
- The output side shall use the same SLAC type (Skarpass) male waveguide flange.

2.2.6 Wall Cooling Circuit

- The accelerating structure shall be provided with eight copper cooling tubes running parallel to the structure axis. The tubes shall be $\frac{1}{2}$ inch outer diameter and shall be grouped in four pairs spaced symmetrically around the periphery of the structure. The feed and return manifold shall be at the middle of the structure so that the cooling system is comprised of eight “U” circuits fed in parallel. The tube shall be slightly flattened where they make contact and shall be brazed longitudinally against the outside of the structure. The water-cooling circuit shall not interfere with the tuning of cavities. Unless otherwise specified, the inlets and outlets to the cooling water manifold shall be

terminated with a stainless steel 3/8 inch schedule 40 male npt thread. The pressure drop from any inlet to outlet shall be approximately 2 PSI for a water flow rate of 1.6 GPM or 13 GPM through all parallel water circuits consecutively. (See attachment)

2.2.7 Disk-loaded Structure

- The disk-loaded structure shall consist of self-aligning cavities brazed together. The cavities shall be of interlocking cup design to reduce the number of braze points. The internal dimensions of cavities shall be graduated to conform to the electrical specifications given in section 2.1. For reference purpose see reference ^[2] SLAC cavity dimensions.

2.2.8 Fabrication Techniques

- All components (except for the possible addition of the external water manifold which at the seller's discretion may be torched brazed in an argon atmosphere to prevent contamination) shall be jointed by brazing in a dry hydrogen furnace. The dew point and temperature shall be such as to avoid copper and chromic oxides. Subsequent handling shall be such that the internal copper surface shall remain shiny and show no evidence of oxidation.
- Generally accepted ultra high vacuum procedures as per reference ^[3] are to be followed while machining, chemically cleaning, brazing and RF tuning. Pieces must be heat treated to relieve stress before final machining. The internal surface shall be finish cut with an approved tool and cutting fluid to achieve a 10 micro-inch or better surface finish before brazing. The outer surface shall have a 25 micro-inch finish. External surfaces of the cups shall be held to a diameter ± 0.001 inches and shall be coaxial with their geometric beam-line axes ± 0.001 inches for alignment purpose.
- Copper shall meet ASTM C10100 specification or better with certification documents to be submitted to the buyer for inspection. The copper for the 86 cavities shall be rough cut to within 0.3 mm of the final dimension, stress relieved in a dry hydrogen furnace at 550 degrees for 40 minutes, inspected and finish machined. No parts with nicks or scratches shall be incorporated in the finished structure.
- Fiduciary provisions for low power laser alignment shall be provided at three locations on the structure, input coupler, middle point, and output coupler.
- After brazing, all surfaces that will be in vacuum shall be handled with gloved hands and with clean, oil free tools and instruments. Great care shall be taken to avoid contamination of the internal surfaces by organic substances or particulate matter of any sort.

2.2.9 Support System

- Support system and "strongback" shall be of non-magnetic material.

- The support systems for the structures shall be such that it would allow installation of quadrupole magnets around the structures. The vendor shall be responsible for the design and construction of the support system. Drawings of the support system shall be approved by the buyer before fabrication.
- In each support structure the accelerating structure shall be supported by a “strongback”. The “strongback” must prevent inelastic deformation of the structure during transport and installation. The structure shall be supported by the “strongback” at four positions, the input coupler, 1/3 point, 2/3 point and the output coupler.
- The support point at the input coupler shall be fixed to the “strongback”. The other three support points shall allow axial differential expansion between the structure and the “strongback”. The support system must accommodate differential expansion when the structure is heated to 250°C for vacuum processing and the “strongback” is at 20°C.
- The “strongback” shall have mounting holes at both ends that shall be used to attach the structure-“strongback” assembly to the buyer’s beamline support structure.

3.0 QUALITY ASSURANCE AND TEST PROCEDURES

3.1 TEST PROCEDURES

The vendor shall inform the buyer at least four weeks in advance as to when the acceptance tests are to be performed so that the buyer may be represented during testing.

3.1.1 Low Power RF Tuning

- Vendor shall specify rf test procedures for review by the buyer.
- Vendor shall provide a list of rf equipment to be used for rf testing and measurements.
- Vendor shall use only currently calibrated equipment to perform rf measurements. Certification of test equipment is to be submitted to the buyer.
- Input and output VSWR of the structure terminated with a matched load versus frequency between 2850 and 2865 MHz for the tuned structure shall be submitted to and inspected by the buyer prior to shipment.
- After tuning, cavity to cavity, cumulative phase shift measurements shall be performed at the intended frequency with compensation for vacuum and temperature (unless measurements are done at operating conditions). The results shall be submitted to the buyer for review prior to shipment.

3.1.2 Fill Time Measurement

- Fill time measurements done at the intended frequency with compensation for vacuum and temperature (unless measurements are done at operating condition) shall be submitted to the buyer for review prior to shipment.

3.1.3 Voltage Attenuation

- Voltage attenuation measurements shall be submitted to the buyer for review before shipment.

3.1.4 Vacuum Leak Tests

- All vacuum leak tests on the finished and tuned structure shall be performed with a liquid nitrogen cold trap between any oil sealed pump and the structure. A cold trap is not required if ion pumping is used. Maximum leak rate as specified in section 2.2.1 shall be demonstrated by the Vendor.

3.1.5 Vacuum Acceptance Test

- After final tuning, the structure shall be vacuum processed until the pressure at the structure end opposite to where pumping (60 l/s) occurs drops to $< 5 \times 10^{-7}$ Torr at the operating temperature of 45°C.

3.1.6 Water Leak Tests

- The water-cooling tubes and manifold shall be statically pressurized at 100 psig for one hour to demonstrate zero leaks. A flow rate measurement shall be carried out to demonstrate pressure drops in the various tubing sections. A flow test shall be performed on each cooling tube circuit to demonstrate that the requirements of section 2.2.6 have been met.

3.1.7 Final Acceptance Tests

- The final acceptance tests including mechanical, vacuum, and rf tests shall be performed at the buyer's site. These tests shall be carried out within 60 days of receipt of each structure. Vendor's representatives are permitted to witness these tests.

3.2 DOCUMENTATION

3.2.1 Milestone

- Three weeks after award of the contract, the vendor shall supply a document listing his major milestones relating to the design, procurement of parts, and fabrication and testing of the cavity cells.

3.2.2 Supplier Proposed Changes

- The supplier shall request all proposed changes in the buyer's approved design, workmanship standards, or manufacturing processes in writing for approval by the buyer prior to implementing the proposed changes.

3.2.3 Nonconformance Reporting

- Any nonconformance to the specification, drawings, or other contract requirements shall be documented as found and submitted to the buyer for disposition prior to further processing.

3.2.4 VSWR Versus Frequency

- A plot of the input and output VSWR versus frequency between 2850 and 2865 MHz for the tuned structure shall be submitted to and approved by the buyer prior to shipping.

3.2.5 Phase Shift

- Measured phase shift data for cavity to cavity (after tuning) and measured cumulative phase shift shall be submitted to and approved by buyer prior to shipping.

3.2.6 Fill Time

- Measured rf fill time at the intended frequency with compensation for vacuum and temperature (unless measurements are done at operating conditions) shall be submitted to and approved by buyer prior to shipping.

3.2.7 Voltage Attenuation

- Voltage attenuation measurements shall be submitted to and approved by buyer prior to shipping.

3.2.8 Vacuum and Water

- Vacuum leak test report listing leak rate, vacuum pressure as per section 3.1.5 and water leak check report including pressure drops shall be submitted to and approved by buyer prior to shipping.

3.2.9 Drawings and Documentation

- All relevant as-built mechanical and electrical drawings necessary to support the equipment installation shall be submitted to buyer with the shipment of each structure.
- All documentation shall conform to the vendor documentation requirements for the Canadian Light Source Inc. ^[4]

3.2.10 Tuning Procedure

- Vendor shall provide the results of RF measurements (Nodal Shift Data).

4.0 DELIVERY REQUIREMENTS

4.1 DELIVERY TIME

- Vendor shall deliver the four structures within eight months after the contract is awarded.

4.2 PACKAGING AND INSURANCE

- Vendor is responsible for the necessary packaging and proper crate to assure safe arrival at buyer location. Also vendor shall be responsible for the delivery insurance.

4.3 STRUCTURE VACUUM PROTECTION

- Structures shall be shipped backfilled with dry nitrogen and “pinched off”.

4.4 PACKAGING AND SHIPPING COST

- Vendor shall be responsible for the cost of packaging and shipping (including transportation insurance).

5.0 MANUFACTURER SPECIFICATIONS

LINAC Cooling/Heating System					
		Section 2	Section 3	Section 4	Section 6
Heater	Manufacturer	Zesta Engineering Ltd	Zesta Engineering Ltd	Zesta Engineering Ltd	Chromalux
	Model	BSC 43303T	BSC 43303T	BSC 43303T	
	Rating	3kW	3kW	3kW	9kW
	Quantity	2	2	2	1
	Heating capacity	6kW	6kW	6kW	9kW
	Set Points	40 °C	40 °C	40 °C	45 °C
Pump	Manufacturer	Goulds Pump	Goulds Pump	Goulds Pump	Goulds Pump
	Model	5GBS0514G4-M003	5GBS0514G4-M003	5GBS0514G4-M003	5GBS0514G4-M003
Motor	HP	0.50	0.50	0.50	0.50
	RPM	3450	3450	3450	3450
	Inlet Pres (psig)	1.5	0.8	1.3	0.4
	Outlet Pres (psig)	11.3	8.3	7.3	7.3
Heat Exchanger	Manufacturer	Alfa-Laval Plate to Plate	Alfa-Laval Plate to Plate	Alfa-Laval Plate to Plate	N/A
	Model	P2-FG	P2-FH	P2-FG	N/A
	Plate Qty	12 x 6mm thick		12 x 6mm thick	N/A
	Plate Area	15.6sqft		15.6sqft	N/A
	Capacity	1.08 US gal		1.08 US gal	N/A
	Hot	1 x 6L		1 x 6L	N/A
	Cold	1 x 5L		1 x 5L	N/A
	Serial #		11421		N/A

6.0 REFERENCES

- [1] ASTM Specs F68-82 (89), B170-89 and B224-87.
- [2] The Stanford Two-Mile Accelerator, p.145, R.B. Neal, 1968.
- [3] Canadian Light Source Vacuum Component Cleaning Technical Procedure, Rev-1A.
- [4] Vendor Documentation Specification, Rev. 3. (0.4.1.1.)